

Business Intelligence — Complete Slip Solutions

USIT6P3 | T.Y.B.Sc. IT — Semester VI | University of Mumbai

Step-by-Step Instructions | Exact Slip Questions | Verified Output

Applications Used in This Exam

Application	Used For
MS SQL Server Management Studio (SSMS)	ETL process, creating databases, staging area, data warehouse
Microsoft Excel	Pivot Table, Pivot Chart, What-If Analysis, importing DW/Cube data
Power BI Desktop	Data visualization, charts, dashboards on sales data
R / RStudio	Classification (time series), K-Means, Linear Regression, Logistic Regression
Spyder IDE (Python)	Pandas CSV analysis, data visualization (matplotlib/seaborn)
SQL Server Data Tools (SSDT / Visual Studio)	Creating OLAP Cube (ROLAP, MOLAP, HOLAP), dimensions, deploying

Practical Index — Cross Reference

Practical	Topic / AIM	Application
1(a)	Import DW data in Excel → Pivot Table & Chart	MS SQL Server + Excel
1(b)	Import OLAP Cube in Excel → Pivot Table & Chart	SSDT + Excel
2	What-If Analysis (Scenario Manager)	Microsoft Excel
3	Classification Algorithm (Time Series)	R / RStudio
4	K-Means Clustering Algorithm	R / RStudio
5	Linear Regression (height vs weight)	R / RStudio
6	Logistic Regression (quality.csv)	R / RStudio
7	CSV Data Analysis using Pandas	Spyder IDE (Python)
8(a)	Data Visualization on Sales Data	Spyder IDE (Python)
8(b)	Data Visualization using Power BI	Power BI Desktop
9	Data Staging Area using SQL	MS SQL Server (SSMS)
10	OLAP Cube – ROLAP/MOLAP/HOLAP	SSDT (Visual Studio) + SSMS

SLIP 1

Max. Marks: 50

Q1 (30 Marks):

- (A) Perform the Extraction Transformation and Loading (ETL) process to construct the database in the SQL Server / Power BI.
- (B) Create the ETL map for the above and setup the schedule for execution.

Q2 (10 Marks):

Show Implementation of Classification algorithm in R Programming.

Question 1 (30 Marks)

"(A) Perform the ETL process to construct the database in SQL Server / Power BI. (B) Create the ETL map and setup the schedule for execution."

Application: Microsoft SQL Server Management Studio (SSMS) | Related To: Practical 9

Step 1: Open SSMS → Connect to server (localhost\SQLEXPRESS)

Step 2: Create CustomerOLTP database and CustOLTP table:

```
CREATE DATABASE CustomerOLTP;
USE CustomerOLTP;

CREATE TABLE CustOLTP (
    custid INT PRIMARY KEY,
    custname VARCHAR(50),
    address VARCHAR(100),
    sal DECIMAL(10,2)
);
```

Step 3: Insert at least 20 records — press F5 to execute:

```
INSERT INTO CustOLTP VALUES
(1, 'Alice', 'Mumbai', 50000), (2, 'Bob', 'Delhi', 60000),
(3, 'Charlie', 'Pune', 45000), (4, 'Diana', 'Nagpur', 55000),
(5, 'Evan', 'Nashik', 70000), (6, 'Fiona', 'Thane', 48000),
(7, 'George', 'Aurangabad', 52000), (8, 'Hannah', 'Kolhapur', 63000),
(9, 'Ivan', 'Solapur', 47000), (10, 'Julia', 'Ahmednagar', 58000),
(11, 'Kevin', 'Latur', 43000), (12, 'Laura', 'Jalgaon', 67000),
(13, 'Mike', 'Nanded', 51000), (14, 'Nina', 'Dhule', 72000),
(15, 'Oscar', 'Satara', 46000), (16, 'Priya', 'Ratnagiri', 59000),
(17, 'Quinn', 'Amravati', 44000), (18, 'Rita', 'Yavatmal', 66000),
(19, 'Sam', 'Beed', 49000), (20, 'Tina', 'Osmanabad', 53000);
```

Step 4: Create CustomerStaging and copy CustOLTP data:

```
CREATE DATABASE CustomerStaging;
USE CustomerStaging;

CREATE TABLE CustStaging (
    custid INT,
    custname VARCHAR(50),
    address VARCHAR(100),
    sal DECIMAL(10,2)
);

INSERT INTO CustStaging
SELECT * FROM CustomerOLTP.dbo.CustOLTP;
```

Step 5: Verify staging data:

```
SELECT * FROM CustomerStaging.dbo.CustStaging;
```

Step 6: Create CustomerDW with transformed columns (ETL transformations):

```
CREATE DATABASE CustomerDW;
USE CustomerDW;

CREATE TABLE CustDW (
    cid          VARCHAR(10),
    cname        VARCHAR(50),
    [add]        VARCHAR(100),
    salINR       DECIMAL(10,2),
    salDollar    DECIMAL(10,2)
);

-- Apply ETL: (i) UPPER case  (ii) salDollar = salINR / 70
INSERT INTO CustDW
SELECT
    UPPER(CAST(custid AS VARCHAR(10))), -- custid in CAPS
    UPPER(custname),                  -- custname in CAPS
    address,
    sal,
    sal / 70                          -- derive salDollar
FROM CustomerStaging.dbo.CustStaging;
```

Step 7: Verify final DW output:

```
SELECT * FROM CustomerDW.dbo.CustDW;
```

VERIFY

cid column shows '1','2' etc. in uppercase format.

cname shows 'ALICE', 'BOB' etc.

salDollar = salINR / 70 (e.g. 50000 / 70 = 714.28)

ETL Map + Schedule — SQL Server Agent

Application: SSMS → SQL Server Agent

Step 1: In SSMS → Expand 'SQL Server Agent' (bottom of Object Explorer) → Right-click 'Jobs' → New Job

Step 2: In General tab: Name the job 'ETL_CustomerJob' → Owner: sa

Step 3: Click 'Steps' tab → New Step → Step Name: 'ETL_Transform' → Type: T-SQL → Database: CustomerDW → Paste the INSERT INTO CustDW query → OK

Step 4: Click 'Schedules' tab → New Schedule → Name: 'Daily_ETL' → Frequency: Daily → Start time: 02:00 AM → OK

Step 5: Click OK to save the job → Right-click the job → 'Start Job at Step' to test

VERIFY

Job runs successfully — green checkmark appears in Job Activity Monitor

Question 2 (10 Marks)

"Show Implementation of Classification algorithm in R Programming."

Step 1: Open RStudio → File → New File → R Script

Step 2: Type and run the following:

```
# Classification using Time Series in R
rainfall <- c(799, 1174.8, 865.1, 1334.6, 635.4, 918.5,
             685.5, 998.6, 784.2, 985, 882.8, 1071)

rainfall.timeseries <- ts(rainfall, start = c(2012, 1), frequency = 12)

print(rainfall.timeseries)
plot(rainfall.timeseries)
```

Step 3: Press Ctrl+A → Ctrl+Enter to run all

VERIFY

Console shows monthly rainfall values Jan 2012 – Dec 2012.

Plots pane shows a line graph: Time on X-axis, Rainfall on Y-axis.

SLIP 2

Max. Marks: 50

Q1 (30 Marks):

- (A) Create the cube with suitable dimension and fact tables based on ROLAP, MOLAP and HOLAP model.
- (B) Import the above cube in Microsoft Excel and create the Pivot table and Pivot Chart to perform data analysis.

Q2 (10 Marks):

Perform the data clustering using clustering algorithm in R Programming.

Question 1 (30 Marks)

"(A) Create the cube with suitable dimension and fact tables based on ROLAP, MOLAP and HOLAP model. (B) Import the above cube in Microsoft Excel and create Pivot Table and Pivot Chart."

PART A — Create the Cube in SSDT (Visual Studio)

Application: SQL Server Data Tools (SSDT / Visual Studio) | Related To: Practical 10

Step 1: Open SQL Server Data Tools (Visual Studio) → Run as Administrator

Step 2: File → New → Project → Business Intelligence → Analysis Services Multidimensional and Data Mining Models → Name: 'SalesOLAP' → OK

Step 3: In Solution Explorer → Right-click 'Data Sources' → New Data Source → Next

Step 4: Click 'New' → Enter Server Name (localhost\SQLEXPRESS) → Select database (Sales_DW or CustomerDW) → Test Connection → OK → Next → 'Inherit' → Next → Finish

Step 5: Right-click 'Data Source Views' → New Data Source View → Select your Data Source → Next

Step 6: Select 'FactProductSales' → '>' → Add Related Tables (adds dimension tables) → Next → Finish

Step 7: Right-click 'Cubes' → New Cube → Next → Select 'Use existing tables' → Next

Step 8: Select FactProductSales as Measure Group Table → Next → Check all measures → Next → Check all Dimensions → Cube name: 'Sales_DW' → Finish

Step 9: Right-click Project → Properties → Deployment → Server Mode: Deploy All → Processing Option: Do Not Process → Server: localhost → OK

Step 10: Right-click Project → Deploy → Wait for 'Deployment Completed' in Output pane

Step 11: Right-click Sales_DW.cube → Process → Run → Wait for 'Process succeeded' → Close

VERIFY

Sales_DW.cube appears in Solution Explorer under Cubes.

Browser tab: drag measures and dimensions to query panel to see data.

ROLAP / MOLAP / HOLAP — Storage Mode Setting

Model	Where to Change	Value
MOLAP	Cube → Partitions tab → Right-click partition → Properties → Storage Mode	MOLAP
ROLAP	Same as above	ROLAP
HOLAP	Same as above	HOLAP

PART B — Import Cube in Excel → Pivot Table & Chart

Application: Microsoft Excel | Related To: Practical 1(b)

Step 1: Open Microsoft Excel → DATA tab → Get External Data → From Other Sources → From Analysis Services

Step 2: Server name: your computer name or localhost → Windows Authentication → Next

Step 3: Select Database: OLAP → Check 'Connect to a specific cube or table' → Select 'Sales DW' → Next → Finish → Save → OK

Step 4: In 'Import Data' dialog → Select 'PivotTable Report' → Existing worksheet → OK

Step 5: PivotTable Fields: drag Product/Customer to ROWS, Time Key to COLUMNS, Quantity/Sales to VALUES

Step 6: Click in Pivot Table → INSERT tab → PivotChart → Column/Bar → OK

VERIFY

PivotTable shows sales data organized by product and time.

PivotChart displays a bar/column chart. Slicers can be added via Analyze tab.

Question 2 (10 Marks)

"Perform the data clustering using clustering algorithm in R Programming."

Application: R / RStudio | Related To: Practical 4

Step 1: Open RStudio → File → New File → R Script

Step 2: Type and run:

```
# K-Means Clustering on Iris Dataset
newiris      <- iris
newiris$Species <- NULL
kc           <- kmeans(newiris, 3)
print(kc)
```

Step 3: Compare species with cluster result:

```
table(iris$Species, kc$cluster)
```

Step 4: Plot clusters and centers:

```
plot(newiris[c('Sepal.Length', 'Sepal.Width')], col = kc$cluster)
points(
  kc$centers[, c('Sepal.Length', 'Sepal.Width')],
  col = 1:3, pch = 8, cex = 2
)
```

Step 5: Ctrl+A → Ctrl+Enter to run entire script

VERIFY

Console: cluster means, clustering vector, within-cluster sum of squares.

Plots: scatter plot with 3 colored clusters and star-shaped centers (pch=8).

SLIP 3

Max. Marks: 50

Q1 (30 Marks):

(A) Create 3 Databases: 1) CustomerOLTP 2) CustomerStaging 3) CustomerDW

1) CustOLTP table (custid, custname, address, sal) — insert 20+ records.

2) CustStaging table in CustomerStaging — copy data from CustOLTP.

3) CustDW table (cid, cname, add, salINR, salDollar). Apply ETL:

(i) custid and custname in CAPITAL CASE.

(ii) Derive salDollar = salINR / 70.

Q2 (10 Marks):

Create the ETL map for the above and setup the schedule for execution.

Question 1 (30 Marks)

"(A) Create 3 Databases — CustomerOLTP, CustomerStaging, CustomerDW — with ETL transformations: (i) custid and custname in CAPITAL CASE (ii) salDollar = salINR / 70"

Application: Microsoft SQL Server Management Studio (SSMS) | Related To: Practical 9

Step 1: Open SSMS → Connect to server (localhost\SQLEXPRESS)

Step 2: Create CustomerOLTP database and CustOLTP table:

```
CREATE DATABASE CustomerOLTP;
USE CustomerOLTP;

CREATE TABLE CustOLTP (
  custid INT PRIMARY KEY,
  custname VARCHAR(50),
  address VARCHAR(100),
  sal DECIMAL(10,2)
);
```

Step 3: Insert at least 20 records — press F5 to execute:

```

INSERT INTO CustOLTP VALUES
  (1, 'Alice', 'Mumbai', 50000),    (2, 'Bob', 'Delhi', 60000),
  (3, 'Charlie', 'Pune', 45000),    (4, 'Diana', 'Nagpur', 55000),
  (5, 'Evan', 'Nashik', 70000),    (6, 'Fiona', 'Thane', 48000),
  (7, 'George', 'Aurangabad', 52000), (8, 'Hannah', 'Kolhapur', 63000),
  (9, 'Ivan', 'Solapur', 47000),    (10, 'Julia', 'Ahmednagar', 58000),
  (11, 'Kevin', 'Latur', 43000),    (12, 'Laura', 'Jalgaon', 67000),
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  (17, 'Quinn', 'Amravati', 44000), (18, 'Rita', 'Yavatmal', 66000),
  (19, 'Sam', 'Beed', 49000),        (20, 'Tina', 'Osmanabad', 53000);

```

Step 4: Create CustomerStaging and copy CustOLTP data:

```

CREATE DATABASE CustomerStaging;
USE CustomerStaging;

CREATE TABLE CustStaging (
  custid INT,
  custname VARCHAR(50),
  address VARCHAR(100),
  sal DECIMAL(10,2)
);

INSERT INTO CustStaging
  SELECT * FROM CustomerOLTP.dbo.CustOLTP;

```

Step 5: Verify staging data:

```

SELECT * FROM CustomerStaging.dbo.CustStaging;

```

Step 6: Create CustomerDW with transformed columns (ETL transformations):

```

CREATE DATABASE CustomerDW;
USE CustomerDW;

CREATE TABLE CustDW (
  cid VARCHAR(10),
  cname VARCHAR(50),
  [add] VARCHAR(100),
  salINR DECIMAL(10,2),
  salDollar DECIMAL(10,2)
);

-- Apply ETL: (i) UPPER case (ii) salDollar = salINR / 70
INSERT INTO CustDW
  SELECT
    UPPER(CAST(custid AS VARCHAR(10))), -- custid in CAPS
    UPPER(custname),                  -- custname in CAPS
    address,
    sal,
    sal / 70                          -- derive salDollar
  FROM CustomerStaging.dbo.CustStaging;

```

Step 7: Verify final DW output:

```

SELECT * FROM CustomerDW.dbo.CustDW;

```

VERIFY

cid column shows '1','2' etc. in uppercase format.

cname shows 'ALICE', 'BOB' etc.

salDollar = salINR / 70 (e.g. 50000 / 70 = 714.28)

Question 2 (10 Marks)

"Create the ETL map for the above and setup the schedule for execution."

ETL Map + Schedule — SQL Server Agent

Application: SSMS → SQL Server Agent

Step 1: In SSMS → Expand 'SQL Server Agent' (bottom of Object Explorer) → Right-click 'Jobs' → New Job

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Step 4: Click 'Schedules' tab → New Schedule → Name: 'Daily_ETL' → Frequency: Daily → Start time: 02:00 AM → OK

Step 5: Click OK to save the job → Right-click the job → 'Start Job at Step' to test

VERIFY

Job runs successfully — green checkmark appears in Job Activity Monitor

SLIP 4

Max. Marks: 50

Q1 (30 Marks):

(A) Create the cube with suitable dimension and fact tables based on ROLAP, MOLAP and HOLAP model.

(B) Import the above cube in Microsoft Excel and create the Pivot table and Pivot Chart to perform data analysis.

Q2 (10 Marks):

Perform the Linear regression on the given data warehouse data.

Question 1 (30 Marks)

"(A) Create the cube (ROLAP, MOLAP, HOLAP). (B) Import cube in Excel → Pivot Table & Pivot Chart."

PART A — Create the Cube in SSDT (Visual Studio)

Application: SQL Server Data Tools (SSDT / Visual Studio) | Related To: Practical 10

Step 1: Open SQL Server Data Tools (Visual Studio) → Run as Administrator

Step 2: File → New → Project → Business Intelligence → Analysis Services Multidimensional and Data Mining Models → Name: 'SalesOLAP' → OK

Step 3: In Solution Explorer → Right-click 'Data Sources' → New Data Source → Next

Step 4: Click 'New' → Enter Server Name (localhost\SQLEXPRESS) → Select database (Sales_DW or CustomerDW) → Test Connection → OK → Next → 'Inherit' → Next → Finish

Step 5: Right-click 'Data Source Views' → New Data Source View → Select your Data Source → Next

Step 6: Select 'FactProductSales' → '>' → Add Related Tables (adds dimension tables) → Next → Finish

Step 7: Right-click 'Cubes' → New Cube → Next → Select 'Use existing tables' → Next

Step 8: Select FactProductSales as Measure Group Table → Next → Check all measures → Next → Check all Dimensions → Cube name: 'Sales_DW' → Finish

Step 9: Right-click Project → Properties → Deployment → Server Mode: Deploy All → Processing Option: Do Not Process → Server: localhost → OK

Step 10: Right-click Project → Deploy → Wait for 'Deployment Completed' in Output pane

Step 11: Right-click Sales_DW.cube → Process → Run → Wait for 'Process succeeded' → Close

VERIFY

Sales_DW.cube appears in Solution Explorer under Cubes.

Browser tab: drag measures and dimensions to query panel to see data.

ROLAP / MOLAP / HOLAP — Storage Mode Setting

Model	Where to Change	Value
MOLAP	Cube → Partitions tab → Right-click partition → Properties → Storage Mode	MOLAP
ROLAP	Same as above	ROLAP
HOLAP	Same as above	HOLAP

PART B — Import Cube in Excel → Pivot Table & Chart

Application: Microsoft Excel | Related To: Practical 1(b)

Step 1: Open Microsoft Excel → DATA tab → Get External Data → From Other Sources → From Analysis Services

Step 2: Server name: your computer name or localhost → Windows Authentication → Next

Step 3: Select Database: OLAP → Check 'Connect to a specific cube or table' → Select 'Sales DW' → Next → Finish → Save → OK

Step 4: In 'Import Data' dialog → Select 'PivotTable Report' → Existing worksheet → OK

Step 5: PivotTable Fields: drag Product/Customer to ROWS, Time Key to COLUMNS, Quantity/Sales to VALUES

Step 6: Click in Pivot Table → INSERT tab → PivotChart → Column/Bar → OK

VERIFY

PivotTable shows sales data organized by product and time.

PivotChart displays a bar/column chart. Slicers can be added via Analyze tab.

Question 2 (10 Marks)

"Perform the Linear regression on the given data warehouse data."

Application: R / RStudio | Related To: Practical 5

Step 1: Open RStudio → File → New File → R Script

Step 2: Enter the following code:

```
# Linear Regression: Height vs Weight
x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)
y <- c( 63,  81,  56,  91,  47,  57,  76,  72,  62,  48)

relation <- lm(y ~ x)
print(relation)
print(summary(relation))
```

Step 3: Predict weight for height = 170:

```
a      <- data.frame(x = 170)
result <- predict(relation, a)
print(result)
```

Step 4: Plot the regression:

```
plot(
  y, x,
  col = 'blue',
  main = 'Height & Weight Regression',
  cex = 1.3, pch = 16,
  xlab = 'Weight in Kg',
  ylab = 'Height in cm'
)
abline(lm(x ~ y))
```

Step 5: Ctrl+A → Ctrl+Enter to run all

VERIFY

Console: Intercept = -38.4551, x coefficient = 0.6746.

Prediction for height 170 = 76.22869.

Plot: blue dots with a fitted regression line.

SLIP 5

Max. Marks: 50

Q1 (30 Marks):

- (A) Perform the Extraction Transformation and Loading (ETL) process to construct the database in the SQL Server.
- (B) Import the data warehouse data in Microsoft Excel and create the Pivot table and Pivot Chart.

Q2 (10 Marks):

Show prediction Using Linear Regression.

Question 1 (30 Marks)

"(A) Perform ETL process to construct the database in SQL Server. (B) Import the DW data in Excel and create Pivot Table and Pivot Chart."

Application: Microsoft SQL Server Management Studio (SSMS) | Related To: Practical 9

Step 1: Open SSMS → Connect to server (localhost\SQLEXPRESS)

Step 2: Create CustomerOLTP database and CustOLTP table:

```
CREATE DATABASE CustomerOLTP;
USE CustomerOLTP;

CREATE TABLE CustOLTP (
  custid  INT PRIMARY KEY,
  custname VARCHAR(50),
  address VARCHAR(100),
  sal     DECIMAL(10,2)
);
```

Step 3: Insert at least 20 records — press F5 to execute:

```
INSERT INTO CustOLTP VALUES
  (1, 'Alice', 'Mumbai', 50000),    (2, 'Bob', 'Delhi', 60000),
  (3, 'Charlie', 'Pune', 45000),    (4, 'Diana', 'Nagpur', 55000),
  (5, 'Evan', 'Nashik', 70000),    (6, 'Fiona', 'Thane', 48000),
  (7, 'George', 'Aurangabad', 52000), (8, 'Hannah', 'Kolhapur', 63000),
  (9, 'Ivan', 'Solapur', 47000),    (10, 'Julia', 'Ahmednagar', 58000),
  (11, 'Kevin', 'Latur', 43000),    (12, 'Laura', 'Jalgaon', 67000),
  (13, 'Mike', 'Nanded', 51000),    (14, 'Nina', 'Dhule', 72000),
  (15, 'Oscar', 'Satara', 46000),    (16, 'Priya', 'Ratnagiri', 59000),
  (17, 'Quinn', 'Amravati', 44000), (18, 'Rita', 'Yavatmal', 66000),
  (19, 'Sam', 'Beed', 49000),       (20, 'Tina', 'Osmanabad', 53000);
```

Step 4: Create CustomerStaging and copy CustOLTP data:

```
CREATE DATABASE CustomerStaging;
USE CustomerStaging;

CREATE TABLE CustStaging (
  custid INT,
  custname VARCHAR(50),
  address VARCHAR(100),
  sal DECIMAL(10,2)
);

INSERT INTO CustStaging
  SELECT * FROM CustomerOLTP.dbo.CustOLTP;
```

Step 5: Verify staging data:

```
SELECT * FROM CustomerStaging.dbo.CustStaging;
```

Step 6: Create CustomerDW with transformed columns (ETL transformations):

```
CREATE DATABASE CustomerDW;
USE CustomerDW;

CREATE TABLE CustDW (
  cid VARCHAR(10),
  cname VARCHAR(50),
  [add] VARCHAR(100),
  salINR DECIMAL(10,2),
  salDollar DECIMAL(10,2)
);

-- Apply ETL: (i) UPPER case  (ii) salDollar = salINR / 70
INSERT INTO CustDW
  SELECT
    UPPER(CAST(custid AS VARCHAR(10))), -- custid in CAPS
    UPPER(custname),                  -- custname in CAPS
    address,
    sal,
    sal / 70                          -- derive salDollar
  FROM CustomerStaging.dbo.CustStaging;
```

Step 7: Verify final DW output:

```
SELECT * FROM CustomerDW.dbo.CustDW;
```

VERIFY

cid column shows '1','2' etc. in uppercase format.

cname shows 'ALICE', 'BOB' etc.

salDollar = salINR / 70 (e.g. 50000 / 70 = 714.28)

Import DW data in Excel → Pivot Table & Chart

Application: Microsoft Excel | Related To: Practical 1(a)

Step 1: Open Microsoft Excel → DATA tab → Get External Data → From Other Sources → From SQL Server

Step 2: Server name: localhost\SQLEXPRESS → Windows Authentication → Next

Step 3: Select database: CustomerDW → Select table CustDW → Next → Finish → Save → OK

Step 4: In 'Import Data' dialog → Select 'PivotTable Report' → OK

Step 5: PivotTable Fields pane: drag cname/address to ROWS, salINR/salDollar to VALUES

Step 6: Click in Pivot Table → INSERT tab → PivotChart → Column or Bar → OK

VERIFY

PivotTable shows DW data grouped by name/address with salary columns.

PivotChart displays a bar/column chart based on the pivot.

Question 2 (10 Marks)

"Show prediction Using Linear Regression."

Application: R / RStudio | Related To: Practical 5

Step 1: Open RStudio → File → New File → R Script

Step 2: Enter the following code:

```
# Linear Regression: Height vs Weight
x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)
y <- c( 63,  81,  56,  91,  47,  57,  76,  72,  62,  48)

relation <- lm(y ~ x)
print(relation)
print(summary(relation))
```

Step 3: Predict weight for height = 170:

```
a      <- data.frame(x = 170)
result <- predict(relation, a)
print(result)
```

Step 4: Plot the regression:

```
plot(
  y, x,
  col = 'blue',
  main = 'Height & Weight Regression',
  cex = 1.3, pch = 16,
  xlab = 'Weight in Kg',
  ylab = 'Height in cm'
)
abline(lm(x ~ y))
```

Step 5: Ctrl+A → Ctrl+Enter to run all

VERIFY

Console: Intercept = -38.4551, x coefficient = 0.6746.

Prediction for height 170 = 76.22869.

Plot: blue dots with a fitted regression line.

SLIP 6

Max. Marks: 50

Q1 (30 Marks):

(A) Create the cube with suitable dimension and fact tables based on ROLAP, MOLAP and HOLAP model.

(B) Execute the MDX queries.

Q2 (10 Marks):

Perform the linear regression on the given data warehouse data.

Question 1 (30 Marks)

"(A) Create the cube (ROLAP, MOLAP, HOLAP). (B) Execute the MDX queries."

PART A — Create the Cube in SSDT (Visual Studio)

Application: SQL Server Data Tools (SSDT / Visual Studio) | Related To: Practical 10

Step 1: Open SQL Server Data Tools (Visual Studio) → Run as Administrator

Step 2: File → New → Project → Business Intelligence → Analysis Services Multidimensional and Data Mining Models → Name: 'SalesOLAP' → OK

Step 3: In Solution Explorer → Right-click 'Data Sources' → New Data Source → Next

Step 4: Click 'New' → Enter Server Name (localhost\SQLEXPRESS) → Select database (Sales_DW or CustomerDW) → Test Connection → OK → Next → 'Inherit' → Next → Finish

Step 5: Right-click 'Data Source Views' → New Data Source View → Select your Data Source → Next

Step 6: Select 'FactProductSales' → '>' → Add Related Tables (adds dimension tables) → Next → Finish

Step 7: Right-click 'Cubes' → New Cube → Next → Select 'Use existing tables' → Next

Step 8: Select FactProductSales as Measure Group Table → Next → Check all measures → Next → Check all Dimensions → Cube name: 'Sales_DW' → Finish

Step 9: Right-click Project → Properties → Deployment → Server Mode: Deploy All → Processing Option: Do Not Process → Server: localhost → OK

Step 10: Right-click Project → Deploy → Wait for 'Deployment Completed' in Output pane

Step 11: Right-click Sales_DW.cube → Process → Run → Wait for 'Process succeeded' → Close

VERIFY

Sales_DW.cube appears in Solution Explorer under Cubes.
Browser tab: drag measures and dimensions to query panel to see data.

ROLAP / MOLAP / HOLAP — Storage Mode Setting

Model	Where to Change	Value
MOLAP	Cube → Partitions tab → Right-click partition → Properties → Storage Mode	MOLAP
ROLAP	Same as above	ROLAP
HOLAP	Same as above	HOLAP

MDX Queries — SSMS (Analysis Services)

Application: SSMS → Connect to Analysis Services | Related To: Practical 10

Step 1: Open SSMS → 'Connect to Server' → Server type: Analysis Services → Server name: localhost → Connect

Step 2: New Query → Select database: OLAP → Write and run MDX:

```
-- Basic MDX: Get all sales by product
SELECT
    [Measures].[Sales Total Cost]      ON COLUMNS,
    [Dim Product].[Product Name].MEMBERS ON ROWS
FROM [Sales Dw]

-- MDX with time filter
SELECT
    [Measures].[Quantity]              ON COLUMNS,
    [Dim Customer].[Customer Name].MEMBERS ON ROWS
FROM [Sales Dw]
WHERE [Dim Time].[Time Key].[2024]
```

Step 3: Press F5 → Results appear in the grid below

VERIFY

Grid shows product names in rows and sales figures in columns.
No error message shown.

Question 2 (10 Marks)

"Perform the linear regression on the given data warehouse data."

Application: R / RStudio | Related To: Practical 5

Step 1: Open RStudio → File → New File → R Script

Step 2: Enter the following code:

```
# Linear Regression: Height vs Weight
x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)
y <- c( 63,  81,  56,  91,  47,  57,  76,  72,  62,  48)

relation <- lm(y ~ x)
print(relation)
print(summary(relation))
```

Step 3: Predict weight for height = 170:

```
a      <- data.frame(x = 170)
result <- predict(relation, a)
print(result)
```

Step 4: Plot the regression:

```
plot(
  y, x,
  col = 'blue',
  main = 'Height & Weight Regression',
  cex = 1.3, pch = 16,
  xlab = 'Weight in Kg',
  ylab = 'Height in cm'
)
abline(lm(x ~ y))
```

Step 5: Ctrl+A → Ctrl+Enter to run all

VERIFY

Console: Intercept = -38.4551, x coefficient = 0.6746.

Prediction for height 170 = 76.22869.

Plot: blue dots with a fitted regression line.

SLIP 7

Max. Marks: 50

Q1 (30 Marks):

- (A) Perform the Extraction Transformation and Loading (ETL) process to construct the database in the SQL Server.
- (B) Apply the What-If Analysis for data visualization. Design and generate necessary reports based on the data warehouse data.

Q2 (10 Marks):

Perform the data clustering using clustering algorithm in R Programming.

Question 1 (30 Marks)

"(A) Perform ETL process to construct the database in SQL Server. (B) Apply What-If Analysis for data visualization and generate reports."

Application: Microsoft SQL Server Management Studio (SSMS) | Related To: Practical 9

Step 1: Open SSMS → Connect to server (localhost\SQLEXPRESS)

Step 2: Create CustomerOLTP database and CustOLTP table:

```
CREATE DATABASE CustomerOLTP;
USE CustomerOLTP;

CREATE TABLE CustOLTP (
  custid  INT PRIMARY KEY,
  custname VARCHAR(50),
  address VARCHAR(100),
  sal     DECIMAL(10,2)
);
```

Step 3: Insert at least 20 records — press F5 to execute:

```
INSERT INTO CustOLTP VALUES
  (1, 'Alice', 'Mumbai', 50000),    (2, 'Bob', 'Delhi', 60000),
  (3, 'Charlie', 'Pune', 45000),    (4, 'Diana', 'Nagpur', 55000),
  (5, 'Evan', 'Nashik', 70000),    (6, 'Fiona', 'Thane', 48000),
  (7, 'George', 'Aurangabad', 52000), (8, 'Hannah', 'Kolhapur', 63000),
  (9, 'Ivan', 'Solapur', 47000),    (10, 'Julia', 'Ahmednagar', 58000),
  (11, 'Kevin', 'Latur', 43000),    (12, 'Laura', 'Jalgaon', 67000),
  (13, 'Mike', 'Nanded', 51000),    (14, 'Nina', 'Dhule', 72000),
  (15, 'Oscar', 'Satara', 46000),    (16, 'Priya', 'Ratnagiri', 59000),
  (17, 'Quinn', 'Amravati', 44000), (18, 'Rita', 'Yavatmal', 66000),
  (19, 'Sam', 'Beed', 49000),       (20, 'Tina', 'Osmanabad', 53000);
```

Step 4: Create CustomerStaging and copy CustOLTP data:

```
CREATE DATABASE CustomerStaging;
USE CustomerStaging;

CREATE TABLE CustStaging (
  custid INT,
  custname VARCHAR(50),
  address VARCHAR(100),
  sal DECIMAL(10,2)
);

INSERT INTO CustStaging
  SELECT * FROM CustomerOLTP.dbo.CustOLTP;
```

Step 5: Verify staging data:

```
SELECT * FROM CustomerStaging.dbo.CustStaging;
```

Step 6: Create CustomerDW with transformed columns (ETL transformations):

```
CREATE DATABASE CustomerDW;
USE CustomerDW;

CREATE TABLE CustDW (
  cid VARCHAR(10),
  cname VARCHAR(50),
  [add] VARCHAR(100),
  salINR DECIMAL(10,2),
  salDollar DECIMAL(10,2)
);

-- Apply ETL: (i) UPPER case  (ii) salDollar = salINR / 70
INSERT INTO CustDW
  SELECT
    UPPER(CAST(custid AS VARCHAR(10))), -- custid in CAPS
    UPPER(custname),                  -- custname in CAPS
    address,
    sal,
    sal / 70                          -- derive salDollar
  FROM CustomerStaging.dbo.CustStaging;
```

Step 7: Verify final DW output:

```
SELECT * FROM CustomerDW.dbo.CustDW;
```

VERIFY

cid column shows '1','2' etc. in uppercase format.

cname shows 'ALICE', 'BOB' etc.

salDollar = salINR / 70 (e.g. 50000 / 70 = 714.28)

What-If Analysis — Scenario Manager

Application: Microsoft Excel | Related To: Practical 2

Step 1: Open Microsoft Excel → Create the data layout:

Cell	Label / Formula	Value
B4	Total books	100
C4	% sold at highest price	0.6 (i.e. 60%)
F10	=B4*C4 (books at highest price)	60
G10	Unit profit (highest)	50
F11	=B4*(1-C4) (books at lowest price)	40
G11	Unit profit (lowest)	20
F12	=F10*G10 + F11*G11 (total revenue)	3800

Step 2: DATA tab → Forecast → What-If Analysis → Scenario Manager

Step 3: Click 'Add' → Scenario name: '60percent' → Changing cells: C4 → OK → Value: 0.6 → OK

Step 4: Repeat Add for: 70percent (0.7), 80percent (0.8), 90percent (0.9), 100percent (1.0)

Step 5: Select scenario → Click 'Show' → Cell F12 updates. Click 'Summary' for a report

VERIFY

60% → Total = 3800 | 70% → Total = 4100

80% → Total = 4400 | 90% → Total = 4700 | 100% → Total = 5000

Question 2 (10 Marks)

"Perform the data clustering using clustering algorithm in R Programming."

Application: R / RStudio | Related To: Practical 4

Step 1: Open RStudio → File → New File → R Script

Step 2: Type and run:

```
# K-Means Clustering on Iris Dataset
newiris <- iris
newiris$Species <- NULL
kc <- kmeans(newiris, 3)
print(kc)
```

Step 3: Compare species with cluster result:

```
table(iris$Species, kc$cluster)
```

Step 4: Plot clusters and centers:

```
plot(newiris[c('Sepal.Length', 'Sepal.Width')], col = kc$cluster)
points(
  kc$centers[, c('Sepal.Length', 'Sepal.Width')],
  col = 1:3, pch = 8, cex = 2
)
```

Step 5: Ctrl+A → Ctrl+Enter to run entire script

VERIFY

Console: cluster means, clustering vector, within-cluster sum of squares.

Plots: scatter plot with 3 colored clusters and star-shaped centers (pch=8).

SLIP 8

Max. Marks: 50

Q1 (30 Marks):

- (A) Create the cube with suitable dimension and fact tables based on ROLAP, MOLAP and HOLAP model.
- (B) Import the above cube in Microsoft Excel and create the Pivot table and Pivot Chart to perform data analysis.

Q2 (10 Marks):

Execute the MDX queries.

Question 1 (30 Marks)

"(A) Create the cube (ROLAP, MOLAP, HOLAP). (B) Import cube in Excel → Pivot Table & Pivot Chart."

PART A — Create the Cube in SSDT (Visual Studio)

Application: SQL Server Data Tools (SSDT / Visual Studio) | Related To: Practical 10

Step 1: Open SQL Server Data Tools (Visual Studio) → Run as Administrator

Step 2: File → New → Project → Business Intelligence → Analysis Services Multidimensional and Data Mining Models → Name: 'SalesOLAP' → OK

Step 3: In Solution Explorer → Right-click 'Data Sources' → New Data Source → Next

Step 4: Click 'New' → Enter Server Name (localhost\SQLEXPRESS) → Select database (Sales_DW or CustomerDW) → Test Connection → OK → Next → 'Inherit' → Next → Finish

Step 5: Right-click 'Data Source Views' → New Data Source View → Select your Data Source → Next

Step 6: Select 'FactProductSales' → '>' → Add Related Tables (adds dimension tables) → Next → Finish

Step 7: Right-click 'Cubes' → New Cube → Next → Select 'Use existing tables' → Next

Step 8: Select FactProductSales as Measure Group Table → Next → Check all measures → Next → Check all Dimensions → Cube name: 'Sales_DW' → Finish

Step 9: Right-click Project → Properties → Deployment → Server Mode: Deploy All → Processing Option: Do Not Process → Server: localhost → OK

Step 10: Right-click Project → Deploy → Wait for 'Deployment Completed' in Output pane

Step 11: Right-click Sales_DW.cube → Process → Run → Wait for 'Process succeeded' → Close

VERIFY

Sales_DW.cube appears in Solution Explorer under Cubes.

Browser tab: drag measures and dimensions to query panel to see data.

ROLAP / MOLAP / HOLAP — Storage Mode Setting

Model	Where to Change	Value
MOLAP	Cube → Partitions tab → Right-click partition → Properties → Storage Mode	MOLAP
ROLAP	Same as above	ROLAP
HOLAP	Same as above	HOLAP

PART B — Import Cube in Excel → Pivot Table & Chart

Application: Microsoft Excel | Related To: Practical 1(b)

Step 1: Open Microsoft Excel → DATA tab → Get External Data → From Other Sources → From Analysis Services

Step 2: Server name: your computer name or localhost → Windows Authentication → Next

Step 3: Select Database: OLAP → Check 'Connect to a specific cube or table' → Select 'Sales DW' → Next → Finish → Save → OK

Step 4: In 'Import Data' dialog → Select 'PivotTable Report' → Existing worksheet → OK

Step 5: PivotTable Fields: drag Product/Customer to ROWS, Time Key to COLUMNS, Quantity/Sales to VALUES

Step 6: Click in Pivot Table → INSERT tab → PivotChart → Column/Bar → OK

VERIFY

PivotTable shows sales data organized by product and time.

PivotChart displays a bar/column chart. Slicers can be added via Analyze tab.

Question 2 (10 Marks)

"Execute the MDX queries."

MDX Queries — SSMS (Analysis Services)

Application: SSMS → Connect to Analysis Services | Related To: Practical 10

Step 1: Open SSMS → 'Connect to Server' → Server type: Analysis Services → Server name: localhost → Connect

Step 2: New Query → Select database: OLAP → Write and run MDX:

```
-- Basic MDX: Get all sales by product
SELECT
    [Measures].[Sales Total Cost]      ON COLUMNS,
    [Dim Product].[Product Name].MEMBERS ON ROWS
FROM [Sales DW]

-- MDX with time filter
SELECT
    [Measures].[Quantity]              ON COLUMNS,
    [Dim Customer].[Customer Name].MEMBERS ON ROWS
FROM [Sales DW]
WHERE [Dim Time].[Time Key].[2024]
```

Step 3: Press F5 → Results appear in the grid below

VERIFY

Grid shows product names in rows and sales figures in columns.

No error message shown.

SLIP 9

Max. Marks: 50

Q1 (30 Marks):

- (A) Create 3 Databases: 1) CustomerOLTP 2) CustomerStaging 3) CustomerDW
- 1) CustOLTP (custid, custname, address, sal) — insert 20+ records.
 - 2) CustStaging — copy data from CustOLTP.
 - 3) CustDW (cid, cname, add, salINR, salDollar). Apply ETL:
 - (i) custid and custname in CAPITAL CASE.
 - (ii) Derive salDollar = salINR / 70.

Q2 (10 Marks):

Perform the data clustering using clustering algorithm.

Question 1 (30 Marks)

"(A) Create 3 Databases — same ETL transformations as Slip 3: (i) CAPITAL CASE (ii) salDollar = salINR / 70"

Application: Microsoft SQL Server Management Studio (SSMS) | Related To: Practical 9

Step 1: Open SSMS → Connect to server (localhost\SQLEXPRESS)

Step 2: Create CustomerOLTP database and CustOLTP table:

```
CREATE DATABASE CustomerOLTP;
USE CustomerOLTP;

CREATE TABLE CustOLTP (
    custid INT PRIMARY KEY,
    custname VARCHAR(50),
    address VARCHAR(100),
    sal DECIMAL(10,2)
);
```

Step 3: Insert at least 20 records — press F5 to execute:

```
INSERT INTO CustOLTP VALUES
(1, 'Alice', 'Mumbai', 50000), (2, 'Bob', 'Delhi', 60000),
(3, 'Charlie', 'Pune', 45000), (4, 'Diana', 'Nagpur', 55000),
(5, 'Evan', 'Nashik', 70000), (6, 'Fiona', 'Thane', 48000),
(7, 'George', 'Aurangabad', 52000), (8, 'Hannah', 'Kolhapur', 63000),
(9, 'Ivan', 'Solapur', 47000), (10, 'Julia', 'Ahmednagar', 58000),
(11, 'Kevin', 'Latur', 43000), (12, 'Laura', 'Jalgaon', 67000),
(13, 'Mike', 'Nanded', 51000), (14, 'Nina', 'Dhule', 72000),
(15, 'Oscar', 'Satara', 46000), (16, 'Priya', 'Ratnagiri', 59000),
(17, 'Quinn', 'Amravati', 44000), (18, 'Rita', 'Yavatmal', 66000),
(19, 'Sam', 'Beed', 49000), (20, 'Tina', 'Osmanabad', 53000);
```

Step 4: Create CustomerStaging and copy CustOLTP data:

```

CREATE DATABASE CustomerStaging;
USE CustomerStaging;

CREATE TABLE CustStaging (
    custid INT,
    custname VARCHAR(50),
    address VARCHAR(100),
    sal DECIMAL(10,2)
);

INSERT INTO CustStaging
    SELECT * FROM CustomerOLTP.dbo.CustOLTP;

```

Step 5: Verify staging data:

```
SELECT * FROM CustomerStaging.dbo.CustStaging;
```

Step 6: Create CustomerDW with transformed columns (ETL transformations):

```

CREATE DATABASE CustomerDW;
USE CustomerDW;

CREATE TABLE CustDW (
    cid VARCHAR(10),
    cname VARCHAR(50),
    [add] VARCHAR(100),
    salINR DECIMAL(10,2),
    salDollar DECIMAL(10,2)
);

-- Apply ETL: (i) UPPER case (ii) salDollar = salINR / 70
INSERT INTO CustDW
    SELECT
        UPPER(CAST(custid AS VARCHAR(10))), -- custid in CAPS
        UPPER(custname),                  -- custname in CAPS
        address,
        sal,
        sal / 70                          -- derive salDollar
    FROM CustomerStaging.dbo.CustStaging;

```

Step 7: Verify final DW output:

```
SELECT * FROM CustomerDW.dbo.CustDW;
```

VERIFY

cid column shows '1','2' etc. in uppercase format.

cname shows 'ALICE', 'BOB' etc.

salDollar = salINR / 70 (e.g. 50000 / 70 = 714.28)

Question 2 (10 Marks)

"Perform the data clustering using clustering algorithm."

Application: R / RStudio | Related To: Practical 4

Step 1: Open RStudio → File → New File → R Script

Step 2: Type and run:

```
# K-Means Clustering on Iris Dataset
newiris      <- iris
newiris$Species <- NULL
kc           <- kmeans(newiris, 3)
print(kc)
```

Step 3: Compare species with cluster result:

```
table(iris$Species, kc$cluster)
```

Step 4: Plot clusters and centers:

```
plot(newiris[c('Sepal.Length', 'Sepal.Width')], col = kc$cluster)
points(
  kc$centers[, c('Sepal.Length', 'Sepal.Width')],
  col = 1:3, pch = 8, cex = 2
)
```

Step 5: Ctrl+A → Ctrl+Enter to run entire script

VERIFY

Console: cluster means, clustering vector, within-cluster sum of squares.

Plots: scatter plot with 3 colored clusters and star-shaped centers (pch=8).

SLIP 10

Max. Marks: 50

Q1 (30 Marks):

- (A) Create the Data staging area for the selected database.
- (B) Import the legacy data from different sources such as (Excel) and load in the target system (SQL Server).

Q2 (10 Marks):

- (A) Implement k-means clustering using R Tool.
 - (B) Show prediction Using Linear Regression.
- [Q2 total = 20 Marks]

Question 1 (20 Marks)

"(A) Create the Data staging area for the selected database. (B) Import legacy data from Excel and load into SQL Server."

PART A — Create Data Staging Area

Application: SSMS | Related To: Practical 9

Create the 3-database ETL structure (see Slip 1 / Slip 3 for full steps).

```
-- Quick reference: staging copy command
INSERT INTO CustomerStaging.dbo.CustStaging
SELECT * FROM CustomerOLTP.dbo.CustOLTP;
```

PART B — Import Legacy Data from Excel to SQL Server

Application: SSMS + Microsoft Excel

Step 1: Prepare Excel file: columns matching SQL table (SalesID, ProductName, SalesAmount ...) → Save as .xlsx

Step 2: In SSMS → Right-click target database → Tasks → Import Data

Step 3: Import and Export Wizard → Next

Step 4: Data Source: 'Microsoft Excel' → Browse to .xlsx → Check 'First row has column names' → Next

Step 5: Destination: SQL Server Native Client → Server: localhost → Windows Auth → Database: Sales_Staging → Next

Step 6: Select 'Copy data from one or more tables' → Next → Map source sheet to destination table → Next → Finish

VERIFY

Wizard shows 'Execution was successful' with row count.

Run `SELECT * FROM target_table` in SSMS to confirm data loaded.

Question 2 (20 Marks) — Two parts

"(A) Implement k-means clustering using R Tool. (B) Show prediction Using Linear Regression."

PART A — K-Means Clustering

Application: R / RStudio | Related To: Practical 4

Step 1: Open RStudio → File → New File → R Script

Step 2: Type and run:

```
# K-Means Clustering on Iris Dataset
newiris      <- iris
newiris$Species <- NULL
kc           <- kmeans(newiris, 3)
print(kc)
```

Step 3: Compare species with cluster result:

```
table(iris$Species, kc$cluster)
```

Step 4: Plot clusters and centers:

```
plot(newiris[c('Sepal.Length', 'Sepal.Width')], col = kc$cluster)
points(
  kc$centers[, c('Sepal.Length', 'Sepal.Width')],
  col = 1:3, pch = 8, cex = 2
)
```

Step 5: Ctrl+A → Ctrl+Enter to run entire script

VERIFY

Console: cluster means, clustering vector, within-cluster sum of squares.

Plots: scatter plot with 3 colored clusters and star-shaped centers (pch=8).

PART B — Linear Regression

Application: R / RStudio | Related To: Practical 5

Step 1: Open RStudio → File → New File → R Script

Step 2: Enter the following code:

```
# Linear Regression: Height vs Weight
x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)
y <- c( 63,  81,  56,  91,  47,  57,  76,  72,  62,  48)

relation <- lm(y ~ x)
print(relation)
print(summary(relation))
```

Step 3: Predict weight for height = 170:

```
a      <- data.frame(x = 170)
result <- predict(relation, a)
print(result)
```

Step 4: Plot the regression:

```
plot(
  y, x,
  col = 'blue',
  main = 'Height & Weight Regression',
  cex = 1.3, pch = 16,
  xlab = 'Weight in Kg',
  ylab = 'Height in cm'
)
abline(lm(x ~ y))
```

Step 5: Ctrl+A → Ctrl+Enter to run all

VERIFY

Console: Intercept = -38.4551, x coefficient = 0.6746.

Prediction for height 170 = 76.22869.

Plot: blue dots with a fitted regression line.

SLIP 11

Max. Marks: 50

Q1 (30 Marks):

(A) Perform the Extraction Transformation and Loading (ETL) process to construct the database in the SQL Server.

Q2 (10 Marks):

(A) Implement k-means clustering using R Tool.
(B) Show prediction Using Linear Regression.

[Q2 total = 20 Marks]

Question 1 (20 Marks)

"(A) Perform the ETL process to construct the database in SQL Server."

Application: Microsoft SQL Server Management Studio (SSMS) | Related To: Practical 9

Step 1: Open SSMS → Connect to server (localhost\SQLEXPRESS)

Step 2: Create CustomerOLTP database and CustOLTP table:

```

CREATE DATABASE CustomerOLTP;
USE CustomerOLTP;

CREATE TABLE CustOLTP (
    custid    INT PRIMARY KEY,
    custname  VARCHAR(50),
    address   VARCHAR(100),
    sal       DECIMAL(10,2)
);

```

Step 3: Insert at least 20 records — press F5 to execute:

```

INSERT INTO CustOLTP VALUES
    (1, 'Alice', 'Mumbai', 50000),    (2, 'Bob', 'Delhi', 60000),
    (3, 'Charlie', 'Pune', 45000),    (4, 'Diana', 'Nagpur', 55000),
    (5, 'Evan', 'Nashik', 70000),     (6, 'Fiona', 'Thane', 48000),
    (7, 'George', 'Aurangabad', 52000), (8, 'Hannah', 'Kolhapur', 63000),
    (9, 'Ivan', 'Solapur', 47000),     (10, 'Julia', 'Ahmednagar', 58000),
    (11, 'Kevin', 'Latur', 43000),     (12, 'Laura', 'Jalgaon', 67000),
    (13, 'Mike', 'Nanded', 51000),     (14, 'Nina', 'Dhule', 72000),
    (15, 'Oscar', 'Satara', 46000),     (16, 'Priya', 'Ratnagiri', 59000),
    (17, 'Quinn', 'Amravati', 44000),  (18, 'Rita', 'Yavatmal', 66000),
    (19, 'Sam', 'Beed', 49000),        (20, 'Tina', 'Osmanabad', 53000);

```

Step 4: Create CustomerStaging and copy CustOLTP data:

```

CREATE DATABASE CustomerStaging;
USE CustomerStaging;

CREATE TABLE CustStaging (
    custid    INT,
    custname  VARCHAR(50),
    address   VARCHAR(100),
    sal       DECIMAL(10,2)
);

INSERT INTO CustStaging
    SELECT * FROM CustomerOLTP.dbo.CustOLTP;

```

Step 5: Verify staging data:

```

SELECT * FROM CustomerStaging.dbo.CustStaging;

```

Step 6: Create CustomerDW with transformed columns (ETL transformations):

```

CREATE DATABASE CustomerDW;
USE CustomerDW;

CREATE TABLE CustDW (
    cid          VARCHAR(10),
    cname        VARCHAR(50),
    [add]        VARCHAR(100),
    salINR       DECIMAL(10,2),
    salDollar    DECIMAL(10,2)
);

-- Apply ETL: (i) UPPER case  (ii) salDollar = salINR / 70
INSERT INTO CustDW
    SELECT
        UPPER(CAST(custid AS VARCHAR(10))), -- custid in CAPS
        UPPER(custname),                  -- custname in CAPS
        address,
        sal,
        sal / 70                          -- derive salDollar
    FROM CustomerStaging.dbo.CustStaging;

```

Step 7: Verify final DW output:

```
SELECT * FROM CustomerDW.dbo.CustDW;
```

VERIFY

cid column shows '1','2' etc. in uppercase format.

cname shows 'ALICE', 'BOB' etc.

salDollar = salINR / 70 (e.g. 50000 / 70 = 714.28)

Question 2 (20 Marks) — Two parts

"(A) Implement k-means clustering using R Tool. (B) Show prediction Using Linear Regression."

PART A — K-Means Clustering

Application: R / RStudio | Related To: Practical 4

Step 1: Open RStudio → File → New File → R Script

Step 2: Type and run:

```

# K-Means Clustering on Iris Dataset
newiris <- iris
newiris$Species <- NULL
kc <- kmeans(newiris, 3)
print(kc)

```

Step 3: Compare species with cluster result:

```
table(iris$Species, kc$cluster)
```

Step 4: Plot clusters and centers:

```
plot(newiris[c('Sepal.Length', 'Sepal.Width')], col = kc$cluster)
points(
  kc$centers[, c('Sepal.Length', 'Sepal.Width')],
  col = 1:3, pch = 8, cex = 2
)
```

Step 5: Ctrl+A → Ctrl+Enter to run entire script

VERIFY

Console: cluster means, clustering vector, within-cluster sum of squares.

Plots: scatter plot with 3 colored clusters and star-shaped centers (pch=8).

PART B — Linear Regression

Application: R / RStudio | Related To: Practical 5

Step 1: Open RStudio → File → New File → R Script

Step 2: Enter the following code:

```
# Linear Regression: Height vs Weight
x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)
y <- c( 63,  81,  56,  91,  47,  57,  76,  72,  62,  48)

relation <- lm(y ~ x)
print(relation)
print(summary(relation))
```

Step 3: Predict weight for height = 170:

```
a      <- data.frame(x = 170)
result <- predict(relation, a)
print(result)
```

Step 4: Plot the regression:

```
plot(
  y, x,
  col = 'blue',
  main = 'Height & Weight Regression',
  cex = 1.3, pch = 16,
  xlab = 'Weight in Kg',
  ylab = 'Height in cm'
)
abline(lm(x ~ y))
```

Step 5: Ctrl+A → Ctrl+Enter to run all

VERIFY

Console: Intercept = -38.4551, x coefficient = 0.6746.

Prediction for height 170 = 76.22869.

Plot: blue dots with a fitted regression line.

SLIP 12

Max. Marks: 50

Q1 (30 Marks):

- (A) Create the Data staging area for the selected database.
- (B) Perform the Extraction Transformation and Loading (ETL) process to construct the database in the SQL Server.

Q2 (10 Marks):

- Perform the data clustering using clustering algorithm in R Programming.

Question 1 (30 Marks)

"(A) Create the Data staging area for the selected database. (B) Perform the ETL process to construct the database in SQL Server."

PART A — Create Data Staging Area

Application: SSMS | Related To: Practical 9

Create the 3-database ETL structure (see Slip 1 / Slip 3 for full steps).

```
-- Quick reference: staging copy command
INSERT INTO CustomerStaging.dbo.CustStaging
SELECT * FROM CustomerOLTP.dbo.CustOLTP;
```

PART B — Import Legacy Data from Excel to SQL Server

Application: SSMS + Microsoft Excel

Step 1: Prepare Excel file: columns matching SQL table (SalesID, ProductName, SalesAmount ...) → Save as .xlsx

Step 2: In SSMS → Right-click target database → Tasks → Import Data

Step 3: Import and Export Wizard → Next

Step 4: Data Source: 'Microsoft Excel' → Browse to .xlsx → Check 'First row has column names' → Next

Step 5: Destination: SQL Server Native Client → Server: localhost → Windows Auth → Database: Sales_Staging → Next

Step 6: Select 'Copy data from one or more tables' → Next → Map source sheet to destination table → Next → Finish

VERIFY

Wizard shows 'Execution was successful' with row count.

Run `SELECT * FROM target_table` in SSMS to confirm data loaded.

PART B — Full ETL Process

Application: Microsoft SQL Server Management Studio (SSMS) | Related To: Practical 9

Step 1: Open SSMS → Connect to server (localhost\SQLEXPRESS)

Step 2: Create CustomerOLTP database and CustOLTP table:

```
CREATE DATABASE CustomerOLTP;
USE CustomerOLTP;

CREATE TABLE CustOLTP (
    custid INT PRIMARY KEY,
    custname VARCHAR(50),
    address VARCHAR(100),
    sal DECIMAL(10,2)
);
```

Step 3: Insert at least 20 records — press F5 to execute:

```
INSERT INTO CustOLTP VALUES
  (1, 'Alice', 'Mumbai', 50000),    (2, 'Bob', 'Delhi', 60000),
  (3, 'Charlie', 'Pune', 45000),    (4, 'Diana', 'Nagpur', 55000),
  (5, 'Evan', 'Nashik', 70000),    (6, 'Fiona', 'Thane', 48000),
  (7, 'George', 'Aurangabad', 52000), (8, 'Hannah', 'Kolhapur', 63000),
  (9, 'Ivan', 'Solapur', 47000),    (10, 'Julia', 'Ahmednagar', 58000),
  (11, 'Kevin', 'Latur', 43000),    (12, 'Laura', 'Jalgaon', 67000),
  (13, 'Mike', 'Nanded', 51000),    (14, 'Nina', 'Dhule', 72000),
  (15, 'Oscar', 'Satara', 46000),    (16, 'Priya', 'Ratnagiri', 59000),
  (17, 'Quinn', 'Amravati', 44000), (18, 'Rita', 'Yavatmal', 66000),
  (19, 'Sam', 'Beed', 49000),       (20, 'Tina', 'Osmanabad', 53000);
```

Step 4: Create CustomerStaging and copy CustOLTP data:

```
CREATE DATABASE CustomerStaging;
USE CustomerStaging;

CREATE TABLE CustStaging (
  custid INT,
  custname VARCHAR(50),
  address VARCHAR(100),
  sal DECIMAL(10,2)
);

INSERT INTO CustStaging
  SELECT * FROM CustomerOLTP.dbo.CustOLTP;
```

Step 5: Verify staging data:

```
SELECT * FROM CustomerStaging.dbo.CustStaging;
```

Step 6: Create CustomerDW with transformed columns (ETL transformations):

```
CREATE DATABASE CustomerDW;
USE CustomerDW;

CREATE TABLE CustDW (
  cid VARCHAR(10),
  cname VARCHAR(50),
  [add] VARCHAR(100),
  salINR DECIMAL(10,2),
  salDollar DECIMAL(10,2)
);

-- Apply ETL: (i) UPPER case (ii) salDollar = salINR / 70
INSERT INTO CustDW
  SELECT
    UPPER(CAST(custid AS VARCHAR(10))), -- custid in CAPS
    UPPER(custname),                  -- custname in CAPS
    address,
    sal,
    sal / 70                          -- derive salDollar
  FROM CustomerStaging.dbo.CustStaging;
```

Step 7: Verify final DW output:

```
SELECT * FROM CustomerDW.dbo.CustDW;
```

VERIFY

cid column shows '1','2' etc. in uppercase format.

cname shows 'ALICE', 'BOB' etc.

salDollar = salINR / 70 (e.g. 50000 / 70 = 714.28)

Question 2 (10 Marks)

"Perform the data clustering using clustering algorithm in R Programming."

Application: R / RStudio | Related To: Practical 4

Step 1: Open RStudio → File → New File → R Script

Step 2: Type and run:

```
# K-Means Clustering on Iris Dataset
newiris      <- iris
newiris$Species <- NULL
kc           <- kmeans(newiris, 3)
print(kc)
```

Step 3: Compare species with cluster result:

```
table(iris$Species, kc$cluster)
```

Step 4: Plot clusters and centers:

```
plot(newiris[c('Sepal.Length', 'Sepal.Width')], col = kc$cluster)
points(
  kc$centers[, c('Sepal.Length', 'Sepal.Width')],
  col = 1:3, pch = 8, cex = 2
)
```

Step 5: Ctrl+A → Ctrl+Enter to run entire script

VERIFY

Console: cluster means, clustering vector, within-cluster sum of squares.

Plots: scatter plot with 3 colored clusters and star-shaped centers (pch=8).

Additional Practicals

Practical 6 — Logistic Regression in R

Application: R / RStudio | Related To: Practical 6 | Dataset: quality.csv

Step 1: Download quality.csv:

```
https://github.com/TarekDib03/Analytics/tree/master/Week3%20-%20Logistic%20Regression/Data
```

Step 2: Open RStudio → New Script → Type:

```
quality <- read.csv('C:/Users/YourName/Downloads/quality.csv')
str(quality)
table(quality$PoorCare)
```

Step 3: Split data into train/test:

```
install.packages('caTools')
library(caTools)
set.seed(88)
split      <- sample.split(quality$PoorCare, SplitRatio = 0.75)
qualityTrain <- subset(quality, split == TRUE)
qualityTest  <- subset(quality, split == FALSE)
```

Step 4: Build logistic regression model:

```
QualityLog <- glm(
  PoorCare ~ OfficeVisits + Narcotics,
  data     = qualityTrain,
  family   = binomial
)
summary(QualityLog)
```

Step 5: Predict and evaluate:

```
predictTrain <- predict(QualityLog, type = 'response')
table(qualityTrain$PoorCare, predictTrain > 0.5)
```

Step 6: Plot ROC curve:

```
install.packages('ROCR')
library(ROCR)
ROCRpred <- prediction(predictTrain, qualityTrain$PoorCare)
ROCRperf <- performance(ROCRpred, 'tpr', 'fpr')
plot(ROCRperf, colorize = TRUE, print.cutoffs.at = seq(0, 1, by = 0.1))
```

VERIFY

Summary shows coefficients for OfficeVisits and Narcotics.

Confusion matrix shows TP/FP/TN/FN counts.

ROC curve plotted with colorized thresholds (0.0 to 1.0).

Practical 7 — Pandas CSV Analysis (Spyder IDE)

Application: Spyder IDE (Python) | Related To: Practical 7

Step 1: Open Spyder IDE → File → New File

Step 2: Create Book1.csv: Excel → columns name, Age, Salary → Save As → .csv

Step 3: Install pandas: pip install pandas (Anaconda Prompt)

Step 4: Type and run:

```
import pandas as pd

df = pd.read_csv('Book1.csv')

print('\nFirst 5 rows:');    print(df.head())
print('\nDataset Summary:'); print(df.info())
print('\nBasic Statistics:'); print(df.describe())
print('\nMissing Values:');  print(df.isnull().sum())

if 'Salary' in df.columns:
    print(f'Average Salary: {df["Salary"].mean():.2f}')
if 'Age' in df.columns:
    print(f'Youngest: {df["Age"].min()}    Oldest: {df["Age"].max()}')
```

Step 5: Press F5 → Check IPython console for output

VERIFY

Shows first 5 rows, dtypes, statistics, missing value counts, salary & age stats.

Practical 8(a) — Data Visualization using Python

Application: Spyder IDE (Python) | Related To: Practical 8(a)

Step 1: Install: pip install pandas numpy matplotlib seaborn

Step 2: Open Spyder → New File → Type:

```

import pandas as pd, numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

dates      = pd.date_range(start='2023-01-01', periods=100, freq='D')
categories = ['Electronics', 'Clothing', 'Groceries', 'Furniture']
data = {
    'Date'      : np.random.choice(dates, 200),
    'Category'   : np.random.choice(categories, 200),
    'Sales_Amount': np.random.randint(100, 1000, 200),
    'Units_Sold' : np.random.randint(1, 20, 200),
}
df = pd.DataFrame(data)
df['Date'] = pd.to_datetime(df['Date'])

# Plot 1: Sales Trend
plt.figure(figsize=(12,6))
sns.lineplot(data=df.groupby('Date')['Sales_Amount'].sum().reset_index(),
             x='Date', y='Sales_Amount')
plt.title('Sales Trend Over Time'); plt.xticks(rotation=45); plt.show()

# Plot 2: Bar Chart by Category
plt.figure(figsize=(10,5))
sns.barplot(data=df.groupby('Category',as_index=False)['Sales_Amount'].sum(),
            x='Category', y='Sales_Amount', palette='viridis')
plt.title('Total Sales by Category'); plt.show()

# Plot 3: Distribution
plt.figure(figsize=(10,5))
sns.histplot(df['Sales_Amount'], bins=20, kde=True, color='blue')
plt.title('Distribution of Sales Amounts'); plt.show()

# Plot 4: Scatter Plot
plt.figure(figsize=(10,5))
sns.scatterplot(data=df, x='Units_Sold', y='Sales_Amount',
               hue='Category', palette='coolwarm')
plt.title('Sales Amount vs Units Sold'); plt.show()

```

Step 3: Press F5 → 4 chart windows open sequentially

VERIFY

Line chart (sales trend), Bar chart (category sales),
Histogram (sales distribution with KDE), Scatter plot (units vs sales by category).

Practical 8(b) — Data Visualization using Power BI

Application: Power BI Desktop | Related To: Practical 8(b)

Step 1: Install Power BI Desktop from <https://powerbi.microsoft.com/downloads/>

Step 2: Prepare sample_sales_data.xlsx: columns → Order Date, Product Name, Category, Region, Sales Amount, Profit, Quantity → 10+ rows → Save

Step 3: Power BI → Get Data → Excel → Browse to .xlsx → Select sheet → Load

Step 4: Click 'Report View' (canvas icon on left sidebar)

Step 5: Visualizations pane → Line Chart → Drag Order Date to X-axis, Sales Amount to Y-axis

Step 6: Click blank canvas → Clustered Bar Chart → Category to Y-axis, Sales Amount to X-axis

Step 7: Filters pane → Drag Region → Select regions to filter

Step 8: Format: click chart → Format tab → add title, change colors, add data labels

Step 9: File → Save → Name: 'SalesDashboard.pbix'

VERIFY

Report shows 2+ interactive charts.

Clicking one chart cross-filters the other. Slicers/filters respond correctly.

Viva Preparation — Key Questions & Answers

ETL & Data Warehouse

Question	Answer
What is ETL?	Extract (get data from source), Transform (clean/convert data), Load (put into DW). Moves data from source to target.
Difference: OLTP vs OLAP?	OLTP: transactional, row-based, daily operations. OLAP: analytical, column-based, supports complex queries.
What is a Staging Area?	Intermediate database where raw data lands before transformation. Buffer between source and target.
What is ROLAP?	Relational OLAP – stores data in relational DB tables. Uses SQL. Slower but handles large data.
What is MOLAP?	Multidimensional OLAP – stores data in multidimensional cubes. Faster, pre-aggregated.
What is HOLAP?	Hybrid OLAP – combines ROLAP and MOLAP. Summary in cubes, detail in relational tables.
What is a Fact Table?	Central table in star schema. Contains measurable data (sales, quantity, profit) and FKs to dimension tables.
What is a Dimension Table?	Contains descriptive attributes (product name, customer, region, date) for grouping/filtering.

Machine Learning / R

Question	Answer
What is K-Means Clustering?	Unsupervised algorithm partitioning data into K clusters. Each point belongs to the nearest centroid.
What is Linear Regression?	Finds a linear relationship between X and Y. Formula: $Y = a + bX$. <code>lm()</code> function in R.
What is Logistic Regression?	Binary classification (0/1). Predicts probability via sigmoid function. <code>glm()</code> with family=binomial in R.
What is the ROC curve?	Receiver Operating Characteristic. Plots True Positive Rate vs False Positive Rate. AUC closer to 1 is better.
What is <code>lm()</code> ?	Linear Model function in R. Syntax: <code>lm(y ~ x, data)</code> . Returns intercept and slope coefficients.
What is <code>ts()</code> ?	Time Series function in R. Syntax: <code>ts(data, start, frequency)</code> . Creates a time series object.

All the Best for Your Examination!
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